
Upvest GmbH

Fractional Trading

Risk & Regulatory Framework

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Chapter 1

Executive Summary

Upvest GmbH operates as a B2B investment infrastructure provider, enabling fintechs, neobanks and brokers to offer their retail clients access to capital markets. The planned launch of fractional trading represents a strategic extension of this infrastructure: rather than requiring investors to purchase whole shares, end users of Upvest's B2B clients will be able to invest fixed euro amounts in high-priced equities, receiving a proportional fractional ownership interest in return. This capability is commercially essential for Upvest's B2B clients to remain competitive with direct-to-consumer platforms such as Trade Republic, Scalable Capital and Revolut, all of which already offer fractional share access.

From a regulatory perspective, Upvest is authorised as a Wertpapierinstitut under the Wertpapierinstitutsgesetz (WpIG) and holds a licence for Eigenhandel pursuant to § 2 Abs. 2 Nr. 10c WpIG, as well as for Depotgeschäft pursuant to § 2 Abs. 3 Nr. 1 WpIG. These licences are well-suited to the fractional trading model: Upvest buys and sells whole shares as principal for its own account as a service to B2B counterparties, and administers the resulting fractional entitlements in custody on behalf of end investors. Under German law, fractional shares are treated as genuine fractional co-ownership interests (Bruchteilseigentum) rather than derivative instruments, a classification that significantly simplifies both the legal structure and the reporting obligations.

The prudential framework applicable to Upvest is the Investment Firms Regulation (IFR, EU 2019/2033). Under IFR, own funds requirements are determined through the K-factor approach — a set of activity-based capital charges calibrated to the risks an investment firm poses to its clients, to the market and to itself. For Upvest, the operative K-factors are K-NPR (net position risk on the trading book, using the same CRR standardised market risk formula as banks), K-ASA (assets safeguarded and administered under its Depotgeschäft licence), K-DTF (daily trading flow as principal) and K-CON (concentration risk towards execution counterparties). The bank concepts of ICAAP and ILAAP do not apply; Upvest's equivalent is the Internal Capital Adequacy and Risk Assessment (ICARA) under IFD Article 24 and WpIG § 39, a single document covering both capital and liquidity adequacy and incorporating a mandatory wind-down plan.

The trading book presents a choice between two fundamentally different inventory

management approaches. A simple fixed-threshold algorithm, in which the system replenishes or reduces whole-share inventory when holdings cross pre-defined levels, carries minimal regulatory overhead and does not trigger MiFID II algorithmic trading classification. A dynamic algorithm that exploits real-time spread conditions, market liquidity and intraday price signals offers better capital efficiency and execution quality, but triggers the full set of obligations under RTS 6 (Commission Delegated Regulation EU 2017/589), including BaFin notification, annual self-assessment, mandatory kill-switch infrastructure and per-order audit trail retention. A hybrid approach — fixed thresholds for the inventory decision combined with TWAP-based execution — is recommended as the initial strategy, with migration to a fully dynamic algorithm deferred until the regulatory infrastructure is in place.

The principal risk drivers are position risk on the inventory book, tail scenarios including flash crashes and settlement failures, market liquidity and market impact costs, and spread-crossing costs. Operationally, reconciliation between the internal fractional ledger and the custody position is the most critical single control: any persistent mismatch constitutes both a financial loss and a potential regulatory breach. The proposed go-live timeline targets general availability in Q1 2027, following legal sign-off, Risk Committee approval and successful completion of a B2B pilot in Q4 2026.

Chapter 2

Product Overview and Business Context

2.1 What Is Fractional Trading?

Fractional trading allows an end investor to acquire a fraction of a single share rather than a whole unit. If a share trades at €3,000, an investor with €50 to deploy may purchase approximately 0.0167 shares, receiving a proportionate ownership interest in the underlying security. The commercial rationale is straightforward: democratisation of access (retail investors with small ticket sizes can hold positions in high-priced stocks), euro-based investing (the investor specifies a notional amount rather than a share quantity), and broad portfolio diversification on a modest budget.

2.2 Upvest's B2B Operating Model

Upvest does not interact with end investors directly. Its clients are regulated intermediaries — fintechs, neobanks and brokers — who build retail investment products on top of Upvest's backend infrastructure covering order routing, custody, settlement and compliance. Figure 2.1 illustrates the layered structure.

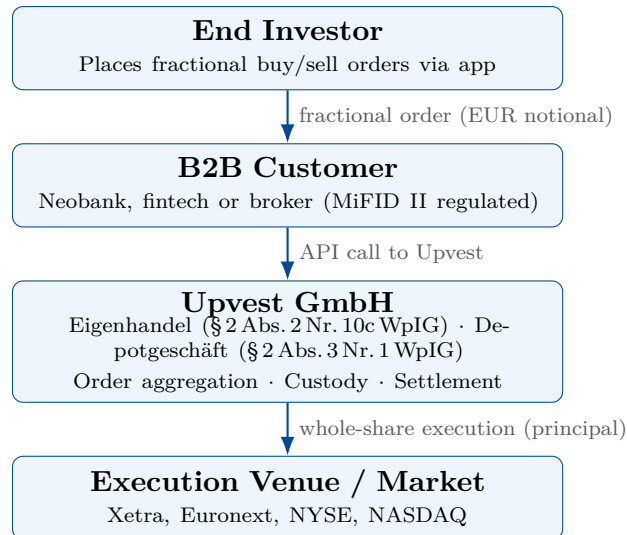


Figure 2.1: Upvest B2B operating model. Upvest acts as principal at every step, bearing aggregation and reconciliation risk.

The critical risk implication of this structure is that Upvest aggregates fractional orders across multiple B2B clients and their end users, executes whole shares in the market as principal, and distributes fractional entitlements internally through its custody ledger. Upvest therefore bears the aggregation risk, the reconciliation risk and the residual position risk; B2B clients bear the relationship with end investors and the associated conduct obligations.

Figure 2.2 shows the order flow from individual fractional instructions through to whole-share execution and pro-rata allocation.

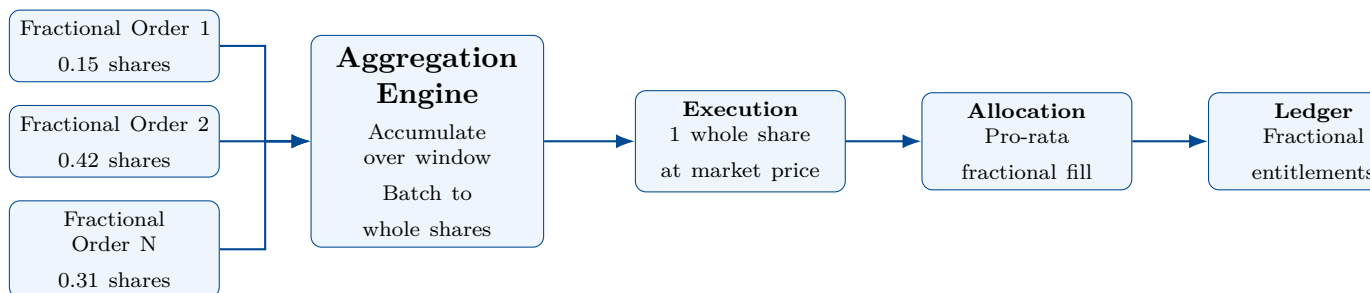


Figure 2.2: Fractional order aggregation and allocation flow. The residual (sub-share remainder) remains on Upvest's trading book.

2.3 Regulatory Context

Upvest operates under German law with BaFin as the primary supervisory authority. The principal EU instruments are MiFID II (conduct) and IFR/IFD (prudential). Fractional shares are classified under German law as genuine fractional co-ownership interests (Bruchteileseigentum), not as derivative instruments. This classification is confirmed by BaFin practice and the 2024 guidance of the French AMF for comparable

structures. Table 2.1 summarises the key regulatory parameters.

Table 2.1: Regulatory context for fractional trading at Upvest

Parameter	Detail
Jurisdiction	Germany (BaFin), European Union (MiFID II / MiFIR, IFR, CSDR)
Instrument classification	Bruchteilseigentum (fractional co-ownership); not a derivative; DepotG applies identically to fractional and whole shares
Custody	CSDR; segregation of client assets required; DepotG compliance
Settlement	T+2 standard; fractional internal bookkeeping must reconcile with whole-share settlement
Corporate actions	Dividends, splits and mergers must be attributed proportionally to fractional holders
Transaction reporting	MiFID II Art. 26 / WpHG § 26: reportable event is the whole-share execution, not individual fractional allocations
Best execution	WpHG § 82 / MiFID II Art. 27: applies to aggregated whole-share executions

Chapter 3

Applicable Regulatory Framework

3.1 Overview of the Framework Stack

As a Wertpapierinstitut, Upvest is not subject to the KWG bank regime. The applicable framework is set by WpIG (primary German law), IFR and IFD (EU prudential requirements, directly applicable), MiFID II / WpHG (conduct), and DORA (digital operational resilience from 17 January 2025). Table 3.1 shows the complete stack.

Table 3.1: Regulatory framework applicable to Upvest

Instrument	Status	Scope for Upvest
WpIG §§ 38–46	In force since 26.06.2021	Primary German law: governance, outsourcing, risk management
IFR (EU) 2019/2033	Directly applicable	K-factors, own funds, liquidity, concentration limits
IFD (EU) 2019/2034	Transposed via WpIG	ICARA (Art. 24), minimum capital (Art. 9), SREP (Art. 36)
MaRisk 06/2024 (BA)	Analogously	Analogous application for small/medium WpIG firms; directly for große Wertpapierfirmen (§ 2 Abs. 18 WpIG)
WpI MaRisk (Konsultation 15/2025)	Under consultation	WpIG-specific risk management circular; expected finalisation end 2025
MaRisk Novelle (Konsultation 02/2026)	Draft	Updated bank MaRisk; monitor for impact on model risk (AT 4.3.4) and outsourcing (AT 9)
WpHG / MiFID II	In force	Best execution, transaction reporting, product governance, algorithmic trading
RTS 6 (EU) 2017/589	If triggered	Algorithmic trading obligations (MiFID II Art. 17); applies to Option B trading algorithm
DORA (EU) 2022/2554	From 17.01.2025	ICT risk framework, incident reporting, TLPT, third-party risk
CRR (EU) 575/2013	K-NPR calc only	K-NPR uses CRR standardised market risk approach (IFR Art. 22); full CRR applies if consolidated assets exceed € 15 bn

MaRisk applicability clarification

MaRisk Rundschreiben 06/2024 (BA) is addressed to Kreditinstitute and Finanzdienstleistungsinstitute under KWG — not to small or medium WpIG Wertpapierinstitute. However, MaRisk AT 2.1 para. 2 explicitly states that *große Wertpapierfirmen* under § 2 Abs. 18 WpIG that are required to apply §§ 25a and 25b KWG must observe MaRisk directly. For small and medium WpIG firms, MaRisk applies analogously as best practice. If Upvest ever reaches the große Wertpapierfirma threshold, MaRisk becomes a direct obligation.

3.2 Eigenhandel — Licence Definition

Upvest’s primary investment service licence is Eigenhandel under § 2 Abs. 2 Nr. 10c WpIG, defined as:

“Anschaffen oder Veräußern von Finanzinstrumenten für eigene Rechnung als Dienstleistung für andere”

(Acquiring or disposing of financial instruments for own account as a service for others.)

This provision directly transposes MiFID II Annex I, Section A, point 3 (dealing on own account), confirmed by IFR Art. 4(9). The defining characteristic is that Upvest acts as **principal** in every transaction — it takes on market risk with its own capital rather than acting as agent for its clients. The service character distinguishes Eigenhandel from pure proprietary Eigengeschäft, which requires no BaFin licence. When Upvest aggregates fractional orders and executes whole shares in the market, it is squarely within Nr. 10c: it acquires financial instruments for its own account as a service rendered to its B2B counterparties. This analysis is further supported by BaFin’s Merkblatt of March 2022 on the distinction between Eigenhandel and Eigengeschäft.

3.3 Permitted and Non-Permitted Activities

The WpIG licence is activity-specific. Table 3.2 sets out the activities Upvest may not conduct without separate authorisation.

Table 3.2: Activities requiring separate licence beyond the current Eigenhandel and Depotgeschäft authorisations

Activity	Legal basis	Note
Accepting deposits	KWG § 32 / WpIG § 15 Abs. 5	Strictly prohibited; WpIG and KWG banking licences cannot be combined
Portfolio management	§ 2 Abs. 2 Nr. 9 WpIG	Discretionary management for clients requires separate authorisation
Agency brokerage (Finanzkommissionsgeschäft)	§ 2 Abs. 2 Nr. 1 WpIG	Eigenhandel always means principal; agency execution is a separate licence
Investment advice	§ 2 Abs. 2 Nr. 4 WpIG	Personalised recommendations require separate licence
Underwriting	§ 2 Abs. 2 Nr. 2 WpIG	Firm commitment or best-efforts placement
MTF / OTF operation	§ 2 Abs. 2 Nr. 6– 7 WpIG	Running a multilateral or organised trading facility
Margin lending / credit	KWG § 1	Requires banking licence

Chapter 4

MaRisk Compliance Framework

4.1 What “MaRisk Compliance” Means for Upvest

The term MaRisk is used loosely in the market, but for Upvest its scope requires careful qualification. BaFin Rundschreiben 06/2024 (BA) — the current and authoritative version of the Mindestanforderungen an das Risikomanagement, dated 29 May 2024 — is formally addressed to Kreditinstitute and Finanzdienstleistungsinstitute under the KWG. As a Wertpapierinstitut under WpIG, Upvest is not a direct addressee of that circular.

Two qualifications apply. First, MaRisk AT 2.1 para. 2 explicitly states that *große Wertpapierfirmen* under § 2 Abs. 18 WpIG — those required by § 4 WpIG to apply §§ 25a and 25b KWG — must observe MaRisk to the extent required by the nature, scale, complexity and risk of their activities. If Upvest ever crosses the *große Wertpapierfirma* threshold, MaRisk 06/2024 (BA) becomes a direct and binding obligation. Second, for small and medium Wertpapierinstitute, MaRisk is applied *sinngemäß* — analogously and as best practice — in the interim pending finalisation of the dedicated **WpI MaRisk** (BaFin Konsultation 15/2025), which is modelled on the MaRisk structure but materially lighter and calibrated to the WpIG framework. The **MaRisk Novelle** (Konsultation 02/2026 (BA)) is a parallel consultation that updates the bank MaRisk and introduces structural changes relevant to Upvest’s forward planning: a new section on supervisory board responsibilities (AT 3.2), a dedicated model risk module (AT 4.3.4 *Verwendung von Modellen*) replacing the prior data management section, and a substantially expanded outsourcing chapter (AT 9).

The operative organisational framework for Upvest today is therefore WpIG §§ 38–46 as primary law, supplemented by MaRisk 06/2024 (BA) applied analogously, DORA from January 2025, and the forthcoming WpI MaRisk once finalised. Table 4.1 summarises the stack.

Table 4.1: MaRisk-related regulatory framework for Upvest

Instrument	Status	Applicability to Upvest
WpIG §§ 38–46	In force	Primary law; organisational requirements directly binding
MaRisk 06/2024 (BA)	In force	Analogous application as best practice; direct for große Wertpapierfirmen
WpI MaRisk (Kons. 15/2025)	Under consultation	WpIG-specific equivalent of MaRisk; expected finalisation end 2025
MaRisk Novelle (Kons. 02/2026)	Draft	Bank MaRisk update; AT 4.3.4 (model risk) and AT 9 (outsourcing) most relevant
DORA (EU) 2022/2554	In force from 17.01.2025	ICT risk, incident reporting, third-party risk; directly applicable
BaFin MaComp (RS 05/2018 (WA))	In force	Conduct and compliance obligations under WpHG

4.2 Governance

4.2.1 Management Board — Vier-Augen-Prinzip

The cornerstone of the WpIG governance framework is the requirement for at least two Geschäftsleiter with sufficient collective knowledge and experience to understand and control all of the firm’s activities (§ 20 WpIG; BaFin Merkblatt 01/2024 (WA)). This requirement is triggered in Upvest’s case not merely by the Eigenhandel licence but by the Depotgeschäft licence and the holding of client assets, both of which independently mandate the two-manager structure. Managers must possess actual, not nominal, understanding of the fractional trading model: they must be able to assess the aggregation mechanism, the principal risk on residual positions and the reconciliation controls. Neither manager may simultaneously serve on the supervisory or administrative board of the same entity (§ 20 Abs. 1 WpIG).

The management board bears full and collective responsibility for the organisational structure, risk strategy, internal control system and risk culture of the firm (MaRisk AT 3). This responsibility cannot be delegated: every material decision about risk appetite, instrument eligibility, inventory thresholds and the choice of trading algorithm must be traceable to the Geschäftsleitung. The MaRisk Novelle formalises this by introducing AT 3.1 Gesamtverantwortung der Geschäftsleitung alongside a new AT 3.2 Verantwortung des Aufsichtsorgans und seiner Ausschüsse, reflecting European supervisory expectations on board-level risk governance that were previously implicit.

4.2.2 Supervisory Body and Committees

Medium-sized Wertpapierinstitute (§ 2 Abs. 17 WpIG) must establish a Risk Committee and a Vergütungskontrollausschuss under § 44 WpIG, unless total assets fall below €100 m (or €300 m under certain conditions). The Risk Committee advises on the

overall risk strategy and must be provided with sufficient information to discharge that function; the remuneration committee reviews incentive structures to ensure they do not create perverse risk-taking incentives. Both bodies feed into the ICARA governance process and must formally record their review of the risk framework at least annually.

4.3 Risk Strategy and Risk Appetite Framework

MaRisk AT 4.2 requires the management board to define and document a **Geschäftsstrategie** (business strategy) and a consistent, derived **Risikostrategie** (risk strategy). The two documents are not independent: the risk strategy must reflect the risk implications of every business decision made in the business strategy. For fractional trading, this means the decision to launch — including the choice of instrument universe, trading algorithm and custody model — must be accompanied by an explicit statement of risk appetite, covering at minimum the maximum permissible residual position per instrument, the ADV participation cap, the reconciliation break tolerance and the overnight inventory policy.

Under MaRisk AT 2.2, the risk inventory — the process by which the firm identifies all material risks — must at minimum consider Adressenausfallrisiken, Marktpreisrisiken, Liquiditätsrisiken and operationelle Risiken. ESG risks must also be integrated into the assessment. For fractional trading the material risks are market risk on the inventory book (Marktpreisrisiken), reconciliation and algorithm failure (operationelle Risiken), settlement counterparty risk (Adressenausfallrisiken) and intraday pre-funding liquidity (Liquiditätsrisiken). The risk strategy document must map each of these to the controls described in this chapter and in the risk framework (Chapter ??).

4.4 Internal Control System

4.4.1 Organisational Structure — Funktionstrennung

MaRisk AT 4.3.1 requires a clear separation of functions such that conflicting activities are not performed by the same person or organisational unit without compensating controls. For the trading book, this principle is developed in detail in BTO 2.1 (Funktionstrennung im Handelsgeschäft), which mandates strict structural separation between the *Handel* (trading / front office) and the *Abwicklung und Kontrolle* (settlement, back office and risk controlling). These functions must report through independent lines to the management board and must not be subordinated to one another.

For Upvest's fractional trading model, the practical demarcation is as follows. The inventory management algorithm and the aggregation engine are front-office functions: they generate orders and determine what trades are executed. The reconciliation process, the settlement monitoring, the K-NPR calculation and the daily P&L attribution are back-office and risk-controlling functions. The same individual may not have authority

over both the order generation logic and the post-trade reconciliation for the same instrument. This separation must exist at the level of system access controls as well as reporting lines.

4.4.2 Risk Controlling and Controlling Processes

MaRisk AT 4.3.2 requires processes for identifying, assessing, managing, monitoring and reporting all material risks. For the trading book, this means an intraday position monitoring process that tracks K-NPR headroom, a daily reconciliation run that validates the fractional ledger against the custody position and a weekly review of K-DTF trends against the rolling average. Limits must be set and documented in the risk strategy, and breaches must trigger a defined escalation process. The risk controlling function must produce regular reporting to the Geschäftsleitung with sufficient granularity to support informed decision-making; BT 3.2 specifies the minimum content of Risikocontrolling reports.

BTO 2.2.3 (*Abbildung im Risikocontrolling*) adds a specific requirement for the trading book: all trading positions must be reflected in the risk controlling system *independently* of the trading system used by the front office. An end-of-day position feed from the execution system to an independent risk ledger, with automated break detection, satisfies this requirement. The independence criterion means the risk controlling function must be able to reconstruct the trading book position from its own data sources, not solely by relying on the front-office system.

4.4.3 Stress Testing

MaRisk AT 4.3.3 requires institutions to conduct regular stress tests covering all material risks. For Upvest's fractional trading book the mandatory programme includes at minimum an adverse scenario applied to the trading inventory (a sudden price fall of 20–40 % on the largest single-instrument position), a correlation stress (simultaneous adverse move across all held instruments), a liquidity stress (normal ADV contracts by 70 % preventing order execution) and an operational stress (reconciliation engine failure for 24 hours). Results must be reviewed by the Geschäftsleitung and feed directly into the ICARA capital adequacy assessment. The MaRisk Novelle substantially expands AT 4.3.3, and Upvest should expect more prescriptive scenario requirements when the WpI MaRisk is finalised.

4.4.4 Model Risk — AT 4.3.4 (MaRisk Novelle)

The MaRisk Novelle (Konsultation 02/2026) replaces the prior data management section with a dedicated **Verwendung von Modellen** module (AT 4.3.4), reflecting the growing supervisory focus on model risk. For Upvest, the most directly affected components are the K-NPR calculation model (which uses the CRR standardised market risk methodology applied to the trading book), the aggregation engine (which determines

batch timing and size), and — if Option B is adopted — the market signal models used to time inventory replenishment. Each model must be inventoried, validated by an independent function before deployment, reviewed periodically, and subject to a change management process that includes re-validation after material changes. Upvest should begin building the model inventory and governance framework in advance of the MaRisk Novelle’s finalisation, particularly if Option B or Option C is the chosen algorithm, since those involve statistical models that will attract immediate scrutiny.

4.5 Three Lines of Defence

MaRisk AT 4.4 specifies three *Besondere Funktionen* that must be organisationally independent of one another and of the business lines they oversee. Table 4.2 sets out the requirements and their specific application to fractional trading.

Table 4.2: Three lines of defence — MaRisk AT 4.4 requirements for Upvest

Function	Requirement	Fractional trading specifics
Risk Controlling (AT 4.4.1)	Independent of trading; direct reporting line to Geschäftsleitung; escalation procedures documented	Must monitor K-NPR intraday, run daily reconciliation, produce weekly K-DTF trend report; cannot be subordinate to the front-office function responsible for the inventory algorithm
Compliance (AT 4.4.2)	Independent of business; BaFin MaComp (RS 05/2018 (WA)) for WpHG conduct obligations; may be combined with risk controlling only if conflicts of interest are excluded	Must sign off on best execution policy, conflicts-of-interest policy and (if applicable) RTS 6 algorithmic trading compliance; must review any new instrument admitted to the fractional universe
Internal Audit (AT 4.4.3 / BT 2)	Full functional separation for medium-sized firms; audit plan covering all material processes; findings reported to Geschäftsleitung	Must audit reconciliation controls, front/back-office segregation, K-factor calculation and outsourcing arrangements; firms with ≤ 10 employees may be fully exempt

4.6 New Product Process — AT 8.1

One of the most directly relevant MaRisk requirements for the fractional trading launch is the **Neu-Produkt-Prozess** mandated by MaRisk AT 8.1. Before a new product or service is offered, or before existing activities are extended into materially new markets or instruments, the institution must complete a structured assessment that covers at

minimum: the risk profile of the new activity and its fit with the existing risk strategy; the required changes to systems, processes and controls; the capital implications; and the legal, compliance and operational readiness of the firm. The Geschäftsleitung must formally approve the new product on the basis of this assessment.

Fractional trading is a new product in the MaRisk AT 8.1 sense: it introduces a new principal dealing model, a new custody obligation for fractional entitlements, new operational processes (aggregation, fractional ledger, corporate action attribution) and new risk exposures (residual position risk, reconciliation failure risk) that do not exist in Upvest's current operations. The AT 8.1 process must be completed and approved before any live fractional order is processed, and the output — the documented risk assessment, systems gap analysis and management sign-off — becomes a core input to the Phase 0/Phase 1 gate in the rollout plan (Chapter ??).

4.7 Trading Book Specific Requirements — BTO 2

For institutions conducting proprietary trading (Handelsgeschäft), MaRisk BTO 2 applies in addition to the general requirements. As the primary circular for trading book governance, BTO 2 translates the general principles of AT 4 into specific organisational and process requirements for the trading function.

4.7.1 Segregation of Duties in the Trading Book (BTO 2.1)

BTO 2.1 reaffirms the strict front/back-office separation for the trading book. The following specific requirements apply to Upvest. Every trade executed by the inventory management algorithm must generate an independent trade confirmation that is processed by the back office without involvement of the front-office function. Positions must be monitored and limits enforced by a function that is organisationally separate from the one managing the algorithm. P&L and mark-to-market valuations must be calculated independently and compared to the front-office valuations on at least a daily basis. Any discrepancy above the reconciliation tolerance must be escalated immediately.

4.7.2 Trading Processes (BTO 2.2)

BTO 2.2.1 (*Handel*) requires that trading activities are conducted within a documented framework of limits, authorities and escalation procedures. For fractional trading this means: documented position limits per instrument and in aggregate; an explicit authority matrix specifying who may change inventory thresholds and with what approval level; and a defined escalation path if a limit is approached or breached. The algorithm itself must operate only within the approved parameter space; any change to thresholds or algorithm logic must follow the change management process defined in AT 4.3.4.

BTO 2.2.2 (*Abwicklung und Kontrolle*) covers settlement and post-trade controls. Every

whole-share execution must generate a settlement instruction that is processed through an independent settlement system, confirmed against the prime broker or venue confirmation, and matched within the standard settlement cycle. Failed settlements must be reported within 24 hours and escalated if unresolved beyond T+3. The reconciliation between the fractional ledger and the custody position at the CSD is the primary BTO 2.2.2 control for fractional trading: it must be automated, run at least daily and produce a timestamped break report that is reviewed by the back-office function independent of the algorithm.

4.8 Outsourcing — § 40 WpIG and MaRisk AT 9

The outsourcing framework under § 40 WpIG requires proportionate safeguards for all material outsourcing arrangements, an Auslagerungsregister covering both material and non-material arrangements, and BaFin notification of material outsourcing via the WpI-AnzV. For a technology-intensive business like fractional trading, the scope of material outsourcing is broad: the execution venue connection, the custody infrastructure, cloud hosting of the aggregation engine, and any third-party data providers feeding the trading algorithm are all potentially material. The overriding principle is that Upvest must retain substantive oversight and cannot delegate core risk management or compliance functions in a way that creates an empty shell.

Third-country outsourcing requires a contractually designated domestic representative (§ 40 Abs. 2 WpIG), and BaFin retains the right to issue orders directly to outsourcing partners (§ 40 Abs. 3 WpIG). The MaRisk Novelle substantially expands AT 9, and the forthcoming WpI MaRisk is expected to address cloud outsourcing, sub-outsourcing chains and exit strategy requirements explicitly. Upvest should treat the current Auslagerungsregister as a living document and ensure every vendor relationship in the fractional trading technology stack is classified correctly before go-live.

4.9 Digital Operational Resilience — DORA

Regulation (EU) 2022/2554 (DORA) has been directly applicable to Wertpapierinstitute since 17 January 2025. It operates alongside the WpIG outsourcing framework and MaRisk AT 9 rather than replacing them: DORA covers ICT-specific risks, while the WpIG and MaRisk frameworks address broader operational and outsourcing risks. The obligations most directly relevant to the fractional trading technology stack are the ICT risk management framework (documented policies and tools for managing ICT risk across the full lifecycle of the aggregation engine and custody platform), the major incident reporting obligation (material ICT-related incidents must be reported to BaFin within defined timeframes), and the ICT third-party risk management framework (mandatory contractual provisions and a register of critical ICT providers). Firms assessed as significant under DORA's proportionality criteria must also conduct threat-

led penetration testing (TLPT); Upvest should assess at go-live whether it falls within that category. Exit strategies for critical ICT dependencies — particularly for cloud providers hosting the aggregation engine — must be documented and tested.

4.10 Compliance Actions — MaRisk Readiness Checklist

Table 4.3 consolidates the MaRisk compliance actions required before and after the fractional trading go-live.

Table 4.3: MaRisk compliance readiness checklist for fractional trading launch

Action	MaRisk reference	Owner / timing
Complete Neu-Produkt-Prozess for fractional trading	AT 8.1	Risk / Compliance; before Phase 1
Document business strategy and risk strategy; include fractional trading risk appetite	AT 4.2	Geschäftsleitung; before Phase 1
Risk inventory: assess all material risks of fractional trading	AT 2.2	Risk; before Phase 1
Establish front/back-office segregation for trading book	BTO 2.1	Operations / Technology; before go-live
Document limits, authority matrix and escalation procedures for trading book	BTO 2.2.1	Risk; before go-live
Implement daily automated reconciliation (fractional ledger vs. custody)	BTO 2.2.2	Technology / Operations; before go-live
Establish independent risk controlling feed from trading system	BTO 2.2.3	Risk Controlling; before go-live
Set up stress testing programme: at minimum 4 scenarios	AT 4.3.3	Risk; before first ICARA
Begin model inventory: list all models in fractional trading stack	AT 4.3.4 (Novelle)	Risk; before go-live
Risk Controlling function: appoint or confirm independence	AT 4.4.1	Geschäftsleitung; before Phase 1

continued ...

Action	MaRisk reference	Owner / timing
Compliance function: update compliance monitoring plan for fractional trading	AT 4.4.2 / MaComp	Compliance; before go-live
Internal Audit: include fractional trading in audit plan	AT 4.4.3 / BT 2	Internal Audit; within 6 months of go-live
Auslagerungsregister: classify all fractional trading vendors	AT 9 / § 40 WpIG	Compliance / Operations; before go-live
Notify BaFin of material outsourcing arrangements	§ 40 WpIG	Compliance; before go-live
DORA: update ICT risk management framework and third-party register	DORA Art. 6, 28	Technology; ongoing
Monitor WpI MaRisk finalisation; assess compliance gap	Kons. 15/2025	Risk; on publication
Monitor MaRisk Novelle; assess impact on model risk (AT 4.3.4) and outsourcing (AT 9)	Kons. 02/2026	Risk; on publication
Monitor große Wertpapierfirma threshold; if reached, MaRisk applies directly	AT 2.1 / § 2 Abs. 18 WpIG	Finance; annual review

Chapter 5

Capital Requirements under IFR

5.1 Framework Structure: IFR versus CRR

Investment firms regulated under IFR are subject to a fundamentally different capital framework from banks. Rather than the risk-weighted-assets (RWA) approach of CRR, IFR employs activity-based K-factors that are calibrated directly to the activities the firm performs. The central design insight is that an investment firm does not take deposits, does not extend credit to the real economy and therefore does not pose the systemic risk that motivates the RWA framework. Table 5.1 compares the two frameworks across key dimensions.

Table 5.1: IFR (Upvest) versus CRR (banks) — key capital framework differences

Dimension	IFR — Upvest	CRR — Banks
Framework logic	Activity-based K-factors	8 % $\times$ risk-weighted assets (RWA)
Market risk	K-NPR = same CRR standardised approach	CRR Title IV Part Three (standardised or FRTB)
Credit risk	Not separately capitalised (no loan book)	Standardised or IRB; 8 % $\times$ RWA
Operational risk	Implicit in FOR (25 % of fixed overheads)	BIA / Standardised / AMA; 8 % $\times$ RWA
Custody / AUM	K-ASA (0.04 %), K-AUM (0.02 %)	Not separately capitalised
Trading volume	K-DTF (0.10 % cash / 0.01 % derivatives)	No direct equivalent
Leverage ratio	Not required	Min. 3 % Tier 1 / total exposure
Liquidity	$\geq \frac{1}{3} \times$ FOR in liquid assets (Art. 43)	LCR (≥ 100 %) + NSFR (≥ 100 %)
Capital minimum	€ 750,000 for Eigenhandel (IFD Art. 9)	€ 5,000,000
ICAAP / ILAAP	Not applicable — replaced by ICARA (IFD Art. 24)	Required under CRD IV Arts. 73, 86
CRR threshold	Full CRR applies if consolidated assets $\geq$ € 15 bn (IFR Art. 1(2))	Always in scope

Key connection between IFR and CRR

K-NPR, the primary market risk capital charge under IFR, uses the **identical formula** as the CRR standardised market risk approach. This is the single most important bridge between the two frameworks: when computing K-NPR, Upvest applies the same equity specific risk (8 % of gross long) and general market risk (8 % of net position) methodology that a bank uses for its trading book.

5.2 Own Funds Requirement — Three-Pillar Structure

Under IFR Art. 11, own funds must at all times equal at least D , defined as:

$$D = \max(R_{\text{FOR}}, K_{\text{min}}, K_{\text{factor}}) \quad (5.1)$$

where R_{FOR} is the Fixed Overhead Requirement, K_{min} is the permanent minimum capital and K_{factor} is the total K-factor requirement. Upvest is **not** a small and non-interconnected investment firm within the meaning of IFR Art. 12(1): dealing on own account means NPR is non-zero, and the Depotgeschäft licence means ASA is non-zero, both of which are disqualifying conditions under Art. 12(1)(c) and (f). All three pillars in equation (5.1) therefore apply.

The Fixed Overhead Requirement is set by IFR Art. 13:

$$R_{\text{FOR}} = \frac{1}{4} \times \text{Fixed Overheads}_{\text{prior year}} \quad (5.2)$$

where fixed overheads exclude variable staff bonuses, profit-sharing, discretionary remuneration, and non-recurring extraordinary items. In the early phase of the launch, when trading volumes are low, R_{FOR} is likely to be the binding pillar: a firm with €12m of annual fixed costs must hold €3m in own funds regardless of K-factor levels. The permanent minimum capital (IFR Art. 14, referencing IFD Art. 9) is €750,000 for firms authorised to deal on own account.

Own funds composition must satisfy (IFR Art. 9):

$$\frac{\text{CET1}}{D} \geq 56\%, \quad \frac{\text{CET1} + \text{AT1}}{D} \geq 75\%, \quad \frac{\text{CET1} + \text{AT1} + \text{T2}}{D} \geq 100\% \quad (5.3)$$

5.3 K-Factor Requirement

The K-factor requirement (IFR Art. 15) is the sum of three risk-to-X categories:

$$K_{\text{factor}} = \underbrace{(K_{\text{ASA}} + K_{\text{CMH}})}_{\text{Risk-to-Client (RtC)}} + \underbrace{K_{\text{NPR}}}_{\text{Risk-to-Market (RtM)}} + \underbrace{(K_{\text{DTF}} + K_{\text{TCD}} + K_{\text{CON}})}_{\text{Risk-to-Firm (RtF)}} \quad (5.4)$$

5.3.1 K-NPR — Net Position Risk (IFR Art. 22)

K-NPR is the primary capital charge for Upvest's trading book, computed using the CRR standardised market risk approach. For each issuer i holding long position L_i and short position S_i (both in EUR notional), and for each FX net position F_{net} :

$$K_{\text{NPR}} = \sum_i \left(0.08 \times L_i + 0.08 \times (L_i - S_i) \right) + 0.08 \times |F_{\text{net}}| \quad (5.5)$$

For a pure long equity inventory (no short hedges), the combined charge is approximately 16% of the gross position per issuer, plus 8% of the net FX exposure. A residual position of €500,000 across several instruments thus attracts roughly €80,000 in K-NPR capital. Three approaches are available under IFR Art. 22: the CRR standardised approach (appropriate for Upvest), the alternative standardised FRTB sensitivities approach, and the alternative internal model approach (requires BaFin approval and is disproportionate for a simple cash equity book).

5.3.2 K-ASA — Assets Safeguarded and Administered (IFR Art. 19)

K-ASA applies because Upvest holds the Depotgeschäft licence and safeguards client fractional positions:

$$K_{\text{ASA}} = 0.04\% \times \overline{\text{ASA}}_{9\text{m}} \quad (5.6)$$

where $\overline{\text{ASA}}_{9\text{m}}$ is the rolling average of daily assets safeguarded over the previous nine months, excluding the three most recent months (IFR Art. 19). At €1 bn of custody AUM, K-ASA contributes €400,000 to own funds requirements; at €5 bn it reaches €2 m.

5.3.3 K-DTF — Daily Trading Flow (IFR Art. 33)

K-DTF captures the operational risk of high-volume principal dealing:

$$K_{\text{DTF}} = 0.10\% \times \overline{\text{DTF}}_{\text{cash}} + 0.01\% \times \overline{\text{DTF}}_{\text{deriv}} \quad (5.7)$$

where DTF is the rolling average of the absolute notional value of daily buys plus sells. As fractional trading scales, K-DTF grows linearly: €100 m of daily cash flow generates

€100,000 of K-DTF capital.

5.3.4 K-TCD — Trading Counterparty Default (IFR Art. 25–26)

K-TCD applies only to OTC derivative contracts, repos and SFTs:

$$K_{\text{TCD}} = 1.2 \times \sum_j \text{EV}_j \times \text{RF}_j \quad (5.8)$$

where $\alpha = 1.2$ is fixed by the Regulation, EV_j is the exposure value and RF_j is the risk factor per counterparty type (0.016 for credit institutions, 0.05 for corporates). For plain cash equity purchases settled DVP, K-TCD is zero. It becomes relevant only if Upvest hedges FX exposure via OTC forwards.

5.3.5 K-CMG — Clearing Margin Given (IFR Art. 23)

K-CMG is an alternative to K-NPR available to firms whose positions are centrally cleared:

$$K_{\text{CMG}} = 1.3 \times M_3 \quad (5.9)$$

where M_3 is the third-highest total margin called on a daily basis by the clearing member over the preceding three months. K-CMG requires explicit BaFin assessment and approval and is only available to firms not belonging to a group containing a credit institution. For Upvest’s cash equity fractional book, K-NPR is the appropriate default approach.

5.3.6 K-CON — Concentration Risk (IFR Art. 35–39)

K-CON applies when a single trading book exposure to an execution counterparty exceeds 25 % of own funds:

$$\text{Limit}_{\text{CON}} = 0.25 \times \text{Own Funds} \quad (5.10)$$

Excess exposure above this limit attracts a progressive capital charge under IFR Art. 39, rising from 2× the excess for the first 40 % band to 9× for excess above 250 % of the limit. Concentration with a single prime broker handling all fractional execution is a realistic K-CON risk: a firm with €8m own funds has a per-counterparty limit of €2m. Multi-venue routing is therefore both a best execution measure and a K-CON capital management tool.

5.3.7 K-factor Summary Table

Table 5.2 summarises all K-factors and their applicability.

Table 5.2: K-factor coefficients (IFR Art. 15, Table 1) and applicability to Upvest

K-factor	Metric	Coefficient	Applies?	Note
K-AUM	Assets under management	0.02 %	No	No portfolio management licence
K-CMH	Client money held (segregated)	0.40 %	TBC	Only if client cash transits Upvest accounts
K-CMH	Client money held (non-seg.)	0.50 %	TBC	Same condition
K-ASA	Assets safeguarded & administered	0.04 %	Yes	Depotgeschäft licence
K-COH	Client orders handled (cash)	0.10 %	No	Only for agent execution
K-DTF	Daily trading flow (cash)	0.10 %	Yes	Eigenhandel
K-DTF	Daily trading flow (derivatives)	0.01 %	If used	OTC hedging only
K-NPR	Net position risk	CRR std. ($\approx 16\%$)	Yes	All trading book positions
K-CMG	Clearing margin given	$1.3 \times M_3$	If approved	Requires BaFin assessment
K-TCD	Trading counterparty default	$1.2 \times \text{EV} \times \text{RF}$	If OTC	OTC derivatives / repos only
K-CON	Concentration (>25 % own funds)	Progressive	Monitor	Per execution counterparty

5.4 Liquidity Requirement

Unlike banks, which must satisfy the Liquidity Coverage Ratio and the Net Stable Funding Ratio, Upvest's liquidity requirement under IFR Art. 43 is straightforward:

$$\text{Liquid Assets} \geq \frac{1}{3} \times R_{\text{FOR}} \quad (5.11)$$

A firm with €3m FOR must therefore hold at least €1m in liquid assets at all times. This is a modest statutory floor; intraday liquidity management for fractional trading — covering pre-funding of client orders, T+2 settlement float and FX settlement — demands considerably more operational liquidity beyond this minimum.

Chapter 6

ICARA — Internal Capital and Liquidity Adequacy

6.1 ICARA Replaces ICAAP and ILAAP

The bank concepts of ICAAP (Internal Capital Adequacy Assessment Process) and ILAAP (Internal Liquidity Adequacy Assessment Process) do not apply to Upvest. In their place, IFD Art. 24, transposed as WpIG § 39, requires an **Internal Capital Adequacy and Risk Assessment (ICARA)** — a single document covering both capital and liquidity adequacy in one proportionate process. Table 6.1 compares the two regimes.

Table 6.1: ICARA (Upvest) versus ICAAP + ILAAP (banks)

	Bank: ICAAP + ILAAP	Upvest: ICARA
Legal basis	CRD IV Arts. 73, 86 / KWG § 25a	IFD Art. 24 / WpIG § 39
Scope	Capital (ICAAP) and liquidity (ILAAP) are separate documents	Combined capital and liquidity assessment
Prescription	Highly prescribed; ECB/BaFin mandated scenario sets	Principles-based; firm designs its own scenarios
Wind-down plan	Not required at the same level	Hard requirement: § 45 Abs. 4 WpIG
Pillar 2 output	P2R (capital add-on) + P2G (guidance)	BaFin SREP (§ 47 WpIG) may impose Pillar 2 add-on
Frequency	Annual minimum	Annual minimum

6.2 ICARA Content Requirements

The ICARA must demonstrate that Upvest continuously holds sufficient capital and liquidity to cover all material risks, remain viable as a going concern and, if needed, execute an orderly wind-down. For a fractional trading business the key elements are as follows.

The **risk inventory** must identify and assess all material risks, including market risk on the trading book inventory and residuals, FX risk from unhedged positions in non-euro instruments, reconciliation and operational risk, custody risk, and concentration risk to execution counterparties. **Capital adequacy** is demonstrated by comparing own funds to the K-factor requirement under both a base case and a stressed scenario: at minimum, a stress that applies a 30 % price drop to the largest single-instrument position and a combined stress that applies a 15 % simultaneous fall across all instruments while assuming an exchange outage prevents unwind for one day. **Liquidity adequacy** covers the IFR Art. 43 liquid assets floor, intraday pre-funding needs and T+2 settlement float. **Business model sustainability** requires a forward-looking capital projection over at least one year that confirms the firm will remain above its own funds requirement as volumes grow.

The **wind-down plan** is a hard statutory requirement under WpIG § 45 Abs. 4 with no direct bank equivalent at the same level of explicitness. It must specify the capital and liquidity needed to execute an orderly exit from the fractional trading business: the time and cost to unwind the residual inventory book, the process for notifying B2B customers and assisting with the transfer or closure of fractional positions, and the settlement obligations that would remain outstanding during the wind-down period.

BaFin reviews the ICARA through the SREP process (WpIG § 47 / IFD Art. 36) and may impose Pillar 2 capital add-ons if the framework is judged inadequate. The ICARA must be approved by the Geschäftsleitung annually.

Chapter 7

Risk Management Framework

7.1 Risk Governance

The risk management framework operates on three lines of defence, illustrated in Figure 7.1. All four risk types required by IFR are addressed within this structure: risks to clients (RtC), risks to the market (RtM), risks to the firm (RtF), and liquidity risk.

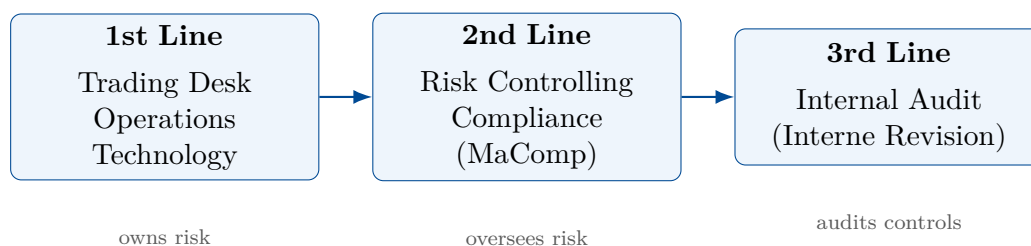


Figure 7.1: Three lines of defence structure. WpIG § 41; MaRisk AT 4.4 (analogously).

7.2 Risk Taxonomy

Table 7.1 presents the risk taxonomy for fractional trading, covering the five categories most relevant to this product.

Table 7.1: Fractional trading risk taxonomy and principal controls

Risk type	Description	Principal controls
Market — position	Price movement on whole-share inventory and sub-share residuals; FX risk on non-EUR instruments	Hard position limits per instrument; intraday K-NPR monitoring; overnight position policy
Market — liquidity	Inability to execute inventory trades at expected prices due to thin market or own market impact	ADV participation cap ($\leq 1\%$ per batch); instrument eligibility floor (min. € 1 m ADV); order slicing
Operational	Reconciliation breaks between fractional ledger and custody position; algorithm failure; settlement fails; data errors	Daily automated reconciliation (± 0.001 share tolerance); kill switch; dual-environment failover; DVP settlement
Credit — counterparty	Prime broker or execution venue default leaving positions unsettled	DVP settlement; K-CON limit (€ = 25 % own funds per counterparty); multi-venue routing
Regulatory	Mis-classification of algorithm as HFT; SI designation; reconciliation breach treated as client asset breach	Legal opinion on algorithm classification; BaFin pre-notification; full RTS 6 compliance for Option B

7.3 Draft Risk Limits

Table 7.2 sets out proposed risk limits for approval by the Risk Committee. These are initial proposals to be calibrated against actual capital levels once the production capital model is complete.

Table 7.2: Draft risk limits for fractional trading book (for Risk Committee approval)

Metric	Limit	Review	Escalation
Max residual notional per instrument	€ 10,000	Daily	Risk
Max intraday K-NPR	80 % of K-NPR capital headroom	Intraday	Risk
ADV participation per batch	1 % of 30d ADV	Per batch	Trading desk
Reconciliation break tolerance	± 0.001 shares per instrument	Daily	Ops → Risk
Max FX open exposure (USD/GBP)	€ 500,000	Daily	Risk
K-CON per execution counterparty	25 % of own funds	Daily	Risk → BaFin
Overnight residual position	€ 50,000 per instrument	Daily at close	Risk

Chapter 8

Trading Book Strategy

8.1 Inventory Management — the Core Design Choice

Upvest's fractional trading model requires a whole-share inventory to fill fractional orders without waiting for full aggregation every time. The algorithm governing when to replenish or reduce that inventory is the central design decision, with significant consequences for regulatory obligations, implementation effort and risk profile. Three options are evaluated.

8.2 Option A — Fixed Threshold Algorithm

The simplest approach sets a minimum, target and maximum inventory level per instrument. When the position falls below the minimum, the system places a market buy to replenish to target. When it exceeds the maximum, it sells to target. No market data feeds into the decision logic; the only input is the current inventory count.

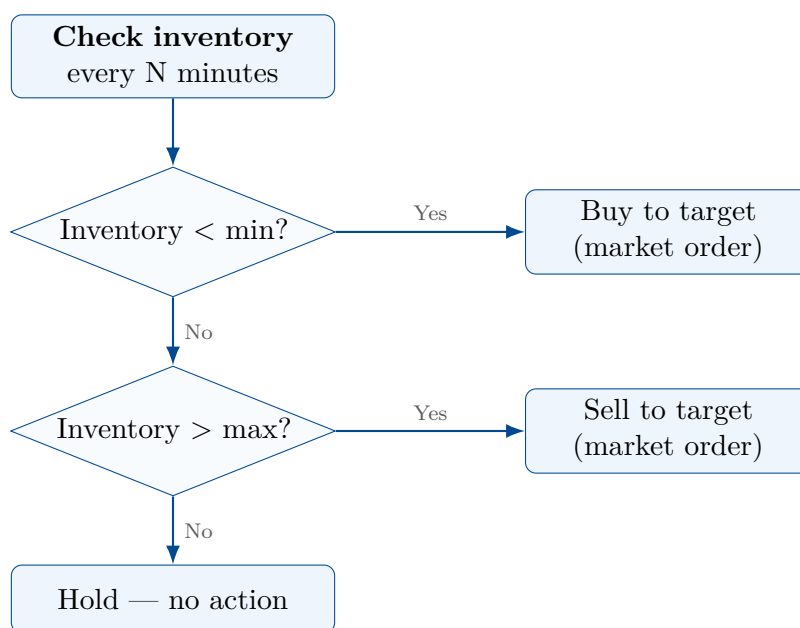


Figure 8.1: Option A decision logic. Example thresholds for a €200 stock: min = 1 share, target = 2, max = 4.

From a regulatory standpoint, Option A is straightforward. Upvest acts as principal and the rule-based automation does not engage market signals to time or size orders; it merely automates what a human trader would do when monitoring inventory levels. This characterisation means algorithmic trading classification under MiFID II Art. 4(1)(39) is unlikely (legal confirmation required), and consequently neither RTS 6 nor WpHG § 80 Abs. 2 notification obligations are triggered. K-NPR is predictable and bounded by the maximum threshold; K-DTF grows modestly with trading volume. The spread cost is borne entirely by Upvest on every replenishment trade, since market orders always cross the bid-ask spread.

8.3 Option B — Dynamic Market-Signal Algorithm

A more advanced algorithm supplements the inventory level signal with real-time market data: bid-ask spread, order-book depth, intraday momentum and a predictive model of fractional order flow by time of day. The system may pre-position inventory ahead of predicted demand peaks, defer replenishment when spreads are wide and attempt to earn the spread by posting limit orders when conditions are favourable.

The commercial benefits are real — tighter spreads, better fill rates and lower average inventory — but the regulatory implications are materially more complex. Automated decision-making that incorporates market signals to determine the timing and size of orders falls squarely within the definition of algorithmic trading in MiFID II Art. 4(1)(39). The full set of RTS 6 obligations applies (detailed in Chapter 9). Additionally, if Upvest systematically quotes prices at which it transacts with B2B clients, a Systematic Internaliser designation under WpIG § 2 Abs. 2 Nr. 10b WpIG may arise, triggering pre-trade transparency obligations. Spread capture by Upvest for its own account creates a conflict of interest that must be disclosed in the conflicts policy and managed to ensure it does not adversely affect fractional order pricing for end investors. The risk of being perceived as front-running client flow — buying inventory moments before a known demand surge — requires careful compliance sign-off and a documented MAR (Market Abuse Regulation) analysis.

8.4 Option C — Recommended Hybrid

Option C uses fixed thresholds for the inventory decision (preserving Option A's regulatory profile) while executing the resulting trades intelligently. Once the system decides to buy or sell, the execution layer slices the order over a defined time window, avoids the first and last fifteen minutes of the session when spreads are widest and sizes each child order to stay within the ADV participation limit. This approach separates the inventory decision (simple, transparent, not a market-signal algorithm) from execution quality (optimised without introducing MAR, conflict of interest or algorithmic trading risks). Legal confirmation is needed that the execution layer does not independently

trigger algorithmic trading classification, but the risk of that is significantly lower than for Option B because the execution logic is purely liquidity-seeking rather than price-signal-seeking. The hybrid is the recommended launch strategy, with migration to Option B deferred until RTS 6 infrastructure is fully built.

Table 8.1 summarises the three options across key dimensions.

Table 8.1: Trading book algorithm options — comparison

Dimension	Option A	Option B	Option C
Logic	Fixed thresholds only	Market signals + inventory level	Fixed decision; smart execution
Algo trading classification	Unlikely	Yes	Confirm with Legal
RTS 6 obligations	Not triggered	Full (see Ch. 9)	Confirm with Legal
BaFin notification	Not required	Required	Confirm with Legal
MAR risk	Minimal	Material	Low
Conflicts of interest	Minimal	Disclosure required	Minimal
K-NPR predictability	High (bounded)	Lower (dynamic)	High
Spread economics	Always taker	Can earn spread	Mostly taker; better timing
Implementation time	1–2 months	6–12 months	2–4 months
Recommended phase	Launch option	Post-stabilisation	Launch (recommended)

Chapter 9

Algorithmic Trading Obligations — RTS 6

9.1 Scope and Trigger

Commission Delegated Regulation (EU) 2017/589 (RTS 6) supplements MiFID II Art. 17 and specifies the organisational requirements for investment firms engaged in algorithmic trading. It is transposed in Germany through WpHG § 80 Abs. 2. The trigger is the use of a computer algorithm that automatically determines individual parameters of orders — whether to initiate, timing, price or quantity — with limited or no human intervention (MiFID II Art. 4(1)(39)). Option B meets this definition; Options A and C require legal assessment as described in the following section.

9.2 The Classification Boundary — ESMA Guidance and BaFin Practice

9.2.1 What ESMA Has Said

ESMA’s Q&A on MiFID II investor protection topics (ESMA35-43-349) addresses algorithmic trading classification through a central principle: the test is not the presence of automation, but whether the computer algorithm is *determining* order parameters versus *transmitting* parameters already determined by a human. A system that merely converts a human instruction into an electronic order — without adding discretion or intelligence to the process — does not by itself constitute algorithmic trading. Pure order routing is the clearest application: if a trader decides to buy 100 shares and uses a system only to send that instruction to a venue, the system is not an algorithm in the MiFID II sense. ESMA has further stated that the assessment must be made case by case; there is no categorical safe harbour for any class of automated system.

This is the foundation of the interpretive argument for Option A. One could characterise a fixed-threshold system as a **standing human instruction**: the human has pre-determined “whenever inventory falls below one share, buy to two shares at market,” and the system merely implements that instruction when the condition arises. On that

reading, the computer is not making a decision; it is executing a decision already made by a human at the time the threshold was programmed.

9.2.2 Why That Argument Is Weaker Than It Looks

Three objections materially limit the standing-instruction argument.

First, Art. 4(1)(39) requires limited or no human intervention at the point the order is generated. A rule set once and then executed automatically without per-order human review is difficult to characterise as involving meaningful human intervention at the point of each individual trade. The human set the rule; the machine applies it to each cycle without any further approval.

Second, the “without adding intelligence” formulation is more demanding than it appears. An Option A system does add something that a pure routing system does not: it determines *whether to initiate* an order on each cycle — inventory below threshold means initiate; otherwise hold. This is a binary decision the machine makes autonomously on every check interval. ESMA’s comfort for routing systems specifically addresses situations where a human has already made the initiation decision; a threshold system is making that decision itself.

Third, ESMA states explicitly that assessment is case by case. No Q&A, guideline or published opinion directly addresses fixed-threshold inventory management in a fractional trading context, and none confirms that such a system falls outside Art. 4(1)(39).

9.2.3 BaFin’s Enforcement Posture

BaFin’s supervisory focus has centred primarily on high-frequency trading, latency-minimising infrastructure and systems that exploit real-time market signals. BaFin has not, to publicly available knowledge, pursued enforcement action against simple automated inventory management systems operating without market data inputs. This enforcement focus creates a practical expectation that a threshold system is less likely to attract scrutiny than a dynamic signal-based algorithm.

However, enforcement posture and legal classification are different things. The absence of enforcement action against threshold systems does not mean BaFin has concluded they fall outside Art. 4(1)(39); it may simply reflect that such systems have not yet come before BaFin in a context requiring a formal determination.

Assessment

The “comfort” for Option A is a plausible interpretive argument grounded in ESMA’s transmit-versus-determine distinction — not a safe harbour. It is strengthened by any human checkpoint in the process, because that supports the limited-human-intervention framing. It weakens materially as soon as the system adds any market data input, execution timing logic or post-submission management. A formal

legal opinion from a MiFID II specialist with direct BaFin supervisory practice is required before relying on this argument in production. A proactive informal BaFin conversation is also advisable: regulators respond significantly better to firms that raise classification questions before launch than after.

9.3 Consequences of BaFin Determining Upvest Is an Algorithmic Trader

If BaFin concludes — whether through proactive engagement, a notification review or a supervisory examination — that Upvest’s inventory management system constitutes algorithmic trading, the consequences fall into two categories: prospective obligations that must be implemented immediately, and retrospective breach consequences for the period during which the system was operated without classification.

9.3.1 Immediate Prospective Obligations

The firm would be required to implement the full RTS 6 framework on an expedited basis. Table 9.1 sets out the key obligations, timelines and implementation complexity.

Table 9.1: RTS 6 remediation obligations upon misclassification finding

Obligation	Timeline	Note
Formal BaFin notification (WpHG § 80 Abs. 2)	Immediate	Administrative; must describe all strategies
Kill switch — build and test	Days to weeks	Must be independent of the algo; accessible to compliance and risk
Pre-trade controls (price collar, order limits, fat-finger)	Weeks	OMS changes; requires testing
Real-time monitoring function	Weeks	Staffing + alerting tooling
Conformance testing at execution venues	4–8 weeks	Requires venue scheduling and CTE access
Per-order audit trail (going forward)	Immediate	Logging infrastructure; gap cannot be cured retroactively
Annual self-assessment framework	Before next review cycle	Process and governance document
Compliance sign-off procedure	Immediate	Governance procedure; low effort

BaFin would typically set a remediation timeline in a formal order (Anordnung). Non-compliance with that order is itself an additional breach, escalating the regulatory exposure.

9.3.2 The Audit Trail Gap — the Irreversible Consequence

The most serious consequence of misclassification is the **retroactive gap in the audit trail**. RTS 6 Art. 9 requires that every order generated by the algorithm carry a unique identifier linking it to the strategy version, parameter set and system state that triggered the decision, retained for five years. If the system was not logging at this level of granularity, that data cannot be reconstructed retroactively. Upvest would have operated, for the entire period from launch, without the record-keeping required by MiFID II Art. 17(2). This is a standalone regulatory breach independent of whether any harm occurred, and it cannot be cured regardless of how quickly everything else is remediated.

9.3.3 Enforcement Risk

The relevant enforcement provision is WpHG § 120, which empowers BaFin to impose administrative fines for violations of algorithmic trading obligations. For legal persons, the maximum fine for violations of WpHG § 80 is:

$$\text{Max fine} = \max(e\,5,000,000, \, 10\% \times \text{total annual turnover}) \quad (9.1)$$

Beyond financial penalties, BaFin may:

- Order the cessation of automated order generation until controls are operational
- Publish the enforcement measure, naming the firm (public naming)
- In cases of persistent or wilful non-compliance, restrict or suspend the Eigenhandel licence
- Impose personal liability on Geschäftsleiter under WpHG § 120 for wilful or negligent violations

9.3.4 Recommended Response Sequence

Upon a BaFin determination of algorithmic trading classification, the response sequence should be executed in the following order.

First, immediately halt automated order submission or accept ongoing breach risk until the kill switch and pre-trade controls are operational — the choice between these two depends on the materiality of the business impact of halting versus the regulatory risk of continued operation. Second, file formal BaFin notification under WpHG § 80 Abs. 2 within the shortest possible timeframe, as continued operation without notification constitutes a separate and ongoing breach. Third, report to the Geschäftsleitung and Risk Committee: board-level awareness is both a governance requirement and evidence of good faith in any subsequent BaFin dialogue. Fourth, engage external legal counsel to assess the full scope of the breach and manage the BaFin relationship. Fifth, reconstruct

the audit trail to the extent possible from available system logs, documenting precisely what data exists and what cannot be recovered. Sixth, implement RTS 6 controls on an emergency basis, prioritising the kill switch and pre-trade controls as the most operationally critical. Finally, consider voluntary disclosure to BaFin of the historical gap before BaFin discovers it independently: regulators consistently treat proactive disclosure as a significant mitigating factor in enforcement decisions, and the contrast between a firm that comes forward and one that waits to be found is substantial in practice.

Why Early Classification Clarity Is Worth the Investment

The cost of a legal opinion and a proactive BaFin conversation before launch is negligible relative to the cost of a retroactive misclassification finding. The audit trail gap alone — five years of non-compliant record-keeping on all automated orders — is a liability that cannot be cured after the fact regardless of remediation speed. The classification decision should be treated as a legal and compliance gate requirement, not a post-launch administrative task.

9.4 Full Obligations under RTS 6

BaFin notification. Before commencing algorithmic trading, the firm must notify BaFin that it is engaged in algorithmic trading and describe the nature of the strategies used. Material changes to strategies require ongoing notification (WpHG § 80 Abs. 2).

Annual self-assessment. RTS 6 Art. 1(2) requires an annual self-assessment of all algorithmic trading activities, reported to the Geschäftsleitung. The assessment must confirm that governance, controls, testing and monitoring remain fit for purpose.

Pre-trade controls (RTS 6 Art. 3). Before any order is submitted, the system must enforce: price collars (block if price deviates beyond a defined range from the reference price); maximum order value and quantity per single submission; maximum position per instrument; a repeated-automated-execution circuit breaker to prevent runaway order loops; and trading halt and tick-size compliance checks.

Post-trade controls (RTS 6 Art. 4). Hard daily loss limits per instrument and firm-wide must automatically halt the algorithm if breached, independent of the trading desk.

Kill switch (RTS 6 Art. 5 / MiFID II Art. 17(1)). A mandatory kill switch must be capable of cancelling all outstanding orders and halting all algorithmic activity immediately, at multiple granularities (per instrument, per strategy, firm-wide). It must operate independently of the algorithm itself and be accessible to compliance and risk functions, not only to the technology team.

Real-time monitoring (RTS 6 Art. 5). A dedicated monitoring function must observe all algorithmic activity during trading hours, detect unusual order patterns and

have a documented escalation procedure with defined response times. This function must be independent of the trading desk.

Testing (RTS 6 Arts. 6–7). Before any deployment and after every significant change to the algorithm, the firm must complete: development environment testing; conformance testing in the execution venue’s conformance testing environment (CTE); stress testing under extreme market conditions; and regression testing confirming that all controls still function. Significant changes include new instruments, new execution venues and changes to the order generation logic.

Compliance sign-off (RTS 6 Art. 8). Compliance must have sufficient understanding of each algorithm to provide effective oversight and must formally sign off before each new strategy is deployed. Annual review is required at minimum.

Audit trail (RTS 6 Art. 9 / MiFID II Art. 17(2)). Every order generated by the algorithm must carry a unique identifier linking it to the specific strategy version, parameter set and system state that triggered the decision. The audit trail must be retained for five years and must be producible to BaFin on request.

Documentation (RTS 6 Art. 10). Required documents include: a full algorithm description covering decision logic, inputs and outputs; a risk and controls document with calibration methodology; a testing log with sign-off by an independent function; a change management log; and the annual self-assessment report.

Table 9.2 shows the implementation roadmap if Option B or Option C is classified as algorithmic trading.

Table 9.2: RTS 6 implementation roadmap for Option B (or Option C if classified)

Phase	Timing	Actions
Pre-development	Before build	Legal classification opinion; BaFin pre-notification discussion; compliance framework design
Development	During build	Pre- and post-trade controls embedded in OMS; kill switch; per-order audit trail
Pre-deployment testing	4–6 weeks before go-live	Development environment test; CTE conformance test at venue; stress test
Regulatory notification	Before go-live	Formal BaFin notification (WpHG § 80 Abs. 2)
Go-live	—	Real-time monitoring active; kill switch tested; compliance sign-off
Year 1 review	12 months post go-live	Annual self-assessment; report to Geschäftsleitung; recalibrate controls

Chapter 10

Trading Book Risk Drivers

10.1 Position Risk

Two distinct exposures contribute to trading book position risk. The first is *planned inventory*: whole shares deliberately held to fill incoming fractional orders before sufficient demand aggregates. In Option A, this is bounded by the maximum threshold; in Option B it is dynamic and can build larger during momentum phases. The second is the *residual position*: the sub-share remainder after fractional allocations are made. If Upvest holds 2.73 shares, allocates 2 to investors and retains 0.73, that 0.73 fraction accumulates continuously across instruments and attracts K-NPR capital of approximately 16 % of its notional value.

The capital implication is direct: a residual position of €500,000 across instruments attracts K-NPR capital of €80,000, with no grace period. Overnight residual positions are particularly important to manage because they cannot be reduced until the next trading session, and adverse price moves overnight affect the full position size. Internal position limits, an explicit overnight position policy and intraday K-NPR monitoring against a capital headroom alert at 80 % are the primary controls.

10.2 Stress

The trading book is exposed to several tail scenarios that the K-NPR standardised charge may not fully capture, and which must therefore be addressed in the ICARA stress testing. A flash crash scenario — where a single instrument falls 20–40 % intraday — creates an immediate mark-to-market loss on the full inventory while demand for fractional orders collapses. A simultaneous correlation shock, where all held instruments fall 15 % on the same day, eliminates any diversification benefit in the book. An exchange outage during the day prevents order execution and forces a larger-than-intended overnight position. Unhedged USD or GBP positions face the compounding risk of both a price fall and an adverse FX move simultaneously. Finally, if a B2B customer cancels a large fractional order after Upvest has pre-positioned inventory, the position must be liquidated into the market without the benefit of matching demand.

Option A provides a hard upper bound on each scenario: the maximum loss equals the

maximum threshold times the stress price move. Option B is potentially unbounded without explicit hard position limits running independently of the algorithm. These hard limits must therefore be treated as a non-negotiable architectural requirement, not as an optional safeguard.

10.3 Market Liquidity and Market Impact

Market liquidity risk arises in two forms. Exogenous liquidity risk reflects the overall depth of the market for a given instrument; during stress periods, available ADV can contract by 60–80 % intraday, making even modestly sized orders difficult to execute without significant market impact. Endogenous risk arises from Upvest’s own orders moving the price against itself. The square-root market impact model provides a useful first-order estimate:

$$\Delta p \approx \sigma \sqrt{\frac{Q}{\text{ADV}}} \quad (10.1)$$

where σ is the daily volatility of the instrument, Q is the order size in EUR and ADV is the 30-day average daily volume. For a €200,000 order in an instrument with 1.5 % daily volatility and €10 m ADV, the estimated impact is approximately $150 \text{ bps} \times \sqrt{0.02} \approx 21 \text{ bps}$ — material enough to require order slicing. An ADV participation cap of 1 % per batch, combined with a minimum ADV eligibility floor of €1 m for any instrument admitted to the fractional trading universe, forms the first line of defence against liquidity risk.

10.4 Spread Crosses

Every time Upvest replenishes or reduces inventory by placing a market order, it crosses the bid-ask spread and absorbs the cost on its own P&L. For a €200 stock with a 5 bps spread, the round-trip cost on a 10-share replenishment is €1. At scale — €100 m of daily fractional flow with 10 round-trip replenishments — the annual spread cost is approximately:

$$\text{Annual spread cost} = \text{Daily flow} \times \text{Spread (bps)} \times \text{Trading days} \approx \text{EUR } 1.25 \text{ m} \quad (10.2)$$

at a 5 bps average spread over 250 trading days. Under stressed market conditions, when spreads widen to 10–15 bps, this cost can double or triple. The spread cost is a direct P&L drag that must be recovered through Upvest’s fee income from B2B clients. Option B’s ability to post limit orders and earn part of the spread improves economics, but introduces adverse selection risk: a limit order that fills only because the price is

moving against Upvest is worse than paying the spread on a market order. The adverse selection rate — the proportion of limit-order fills followed by a price move against Upvest within 30 seconds — is therefore a primary algorithm health KPI for Option B, with a target below 35 %.

10.5 Operational Risk

10.5.1 Reconciliation

Reconciliation failure is the most critical operational risk in the fractional trading model. The internal fractional ledger must at all times match the whole-share position held in custody at the CSD, net of all fractional allocations. Any persistent break means either over-allocation (Upvest has credited more shares to investors than it holds) or under-allocation (investors' entitlements are understated), both of which are financial losses and potential regulatory breaches under the DepotG.

Causes of reconciliation breaks include rounding error accumulation from floating-point arithmetic in fractional calculations, missed corporate action adjustments, settlement fails at the CSD where shares are credited to the fractional ledger before actual receipt and duplicate order processing from system retries. The primary control is a daily automated reconciliation with a tolerance of ± 0.001 shares per instrument; any break exceeding tolerance triggers immediate alert and must be resolved within 24 hours, with escalation to the Risk Committee if unresolved.

10.5.2 Algorithm and System Failure

For Option B and Option C, system failure carries elevated risk. A software bug in the order generation logic can create a runaway order loop, submitting excessive orders before the circuit breaker activates; a model error in the flow prediction or spread estimation can lead to systematically incorrect inventory decisions that accumulate losses before detection; a stale market data feed can cause the algorithm to operate on prices that are several seconds old, leading to incorrect execution. The kill switch (mandatory under RTS6 for Option B) must function independently of the algorithm and be tested on a regular schedule. A heartbeat monitoring process that raises an alert if the aggregation engine misses its check-in within a defined window, and a hot standby environment capable of failover within 60 seconds, form the minimum infrastructure requirements.

10.5.3 Settlement and Counterparty Risk

Settlement failure — where a counterparty delivers shares late or not at all — leaves Upvest holding a position that is not matched by actual custody receipt. DVP (delivery-versus-payment) settlement eliminates the principal component of settlement risk, but

timing mismatches still arise. A prime broker default during the settlement window is the most severe form: positions held in the prime broker's omnibus account enter the insolvency estate. Multi-venue routing and the K-CON limit of 25 % of own funds per counterparty are the primary structural mitigants, supplemented by a daily settlement fail monitoring report with escalation for any failure persisting beyond T+3.

Chapter 11

Rollout Plan

11.1 Phased Approach

The launch is structured in four phases, with clear exit criteria governing the transition between each. Figure 11.1 illustrates the timeline.

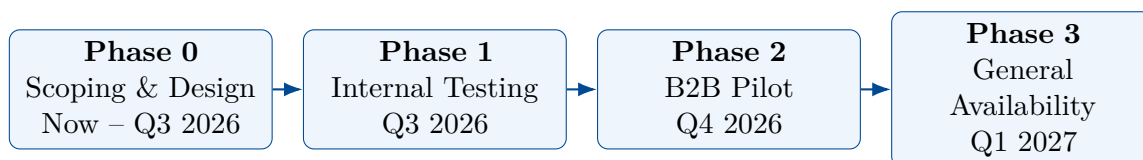


Figure 11.1: Rollout phases and indicative timeline.

Phase 0 covers the decisions that must be made before any technical build begins: confirmation of the instrument universe, legal opinion on the regulatory classification of the custody and aggregation structure, selection of the trading book algorithm (this document provides the framework for that decision), Risk Committee approval of the risk framework, and preliminary engagement with BaFin if required. The exit criterion is legal sign-off and an approved risk framework.

Phase 1 moves into internal testing: the aggregation engine, fractional ledger and reconciliation processes are built and unit tested; corporate action attribution is validated; dry-run reconciliations are run against simulated custody positions. The exit criterion is zero critical defects and demonstrated reconciliation accuracy of $\geq 99.99\%$ across a defined test period.

Phase 2 is a controlled pilot with one or two B2B customers, restricted to a limited instrument universe (for example, the top 50 European equities by liquidity). Daily manual oversight of reconciliation and residual positions is maintained throughout. The exit criterion is four consecutive weeks of clean operation with no material reconciliation breaks and all risk limits respected.

Phase 3 opens fractional trading to all B2B customers with an expanded instrument universe, automated monitoring and alerting fully operational, and risk reporting integrated into the standard management information pack.

11.2 Go-Live Risk Checklist

Table 11.1 shows the actions that must be completed before Phase 3 general availability.

Table 11.1: Go-live risk checklist

Item	Owner	Status
Legal opinion on instrument classification and custody structure	Legal	TBD
Risk framework approved by Risk Committee	Risk	TBD
Best execution policy updated for aggregated orders	Compliance	TBD
B2B customer agreements updated	Legal	TBD
ICARA updated to include fractional trading	Risk	TBD
Wind-down plan documented	Risk	TBD
Reconciliation engine tested and signed off	Technology	TBD
Corporate action attribution validated	Operations	TBD
Kill switch tested (if Option B or C algo trading)	Technology	TBD
BaFin algorithmic trading notification (if triggered)	Compliance	TBD
DORA ICT third-party risk register updated	Technology	TBD
Risk limits loaded and monitored in risk systems	Risk	TBD
Incident runbook signed off	Risk / Operations	TBD

11.3 Open Questions

Several decisions remain outstanding and must be resolved before the Phase 0 exit: the custody model (omnibus account versus segregated per B2B client); the instrument universe for Version 1 (EU equities only, or including US equities with associated FX complexity); the question of whether any client cash transits Upvest's own accounts during the order cycle (determining K-CMH applicability); the legal view on Systematic Internaliser risk for Option B; and BaFin's position on whether the Option C hybrid execution layer independently triggers algorithmic trading classification.

Appendix A

Trading Book Quantitative Analysis

This appendix presents a quantitative analysis of the fractional trading book across two dimensions: order execution throughput and inventory size. The analysis is parametric, covering a price universe of €0.01 to €400 per share, instrument counts of up to 5,000 and daily fractional order flow of €40 million, with a realistic distribution in which 80 % of instruments are priced below €50. All calculations are produced by the Python models in `models/trading_book_simulation.py`, `trading_book_chart.py` and `trading_book_inventory_chart.py`.

A.1 Model Assumptions

The model is built on a closed-form analytical expression. For a universe of N instruments each with average share price P and a total daily fractional order flow of F , the number of whole-share execution events per second is:

$$\text{trades/sec} = \frac{F}{P \times r \times T} \quad (\text{A.1})$$

where r is the replenishment size in shares (the number of whole shares purchased in each inventory replenishment trade, equal to `max_inventory - min_inventory`) and T is the length of the trading session in seconds ($7 \text{ h} = 25,200 \text{ s}$). A counter-intuitive but important result follows directly from equation (A.1): the instrument count N cancels, because increasing the number of instruments distributes the same total flow across more securities, leaving each instrument with proportionally less demand. The throughput constraint is therefore determined entirely by the average price level, the replenishment window width and the daily flow — not by the breadth of the instrument universe. The inventory notional, by contrast, grows linearly with both N and P :

$$\text{avg inventory notional} = N \times \bar{s} \times P \quad (\text{A.2})$$

where $\bar{s} = (\text{max} + \text{min})/2 = 3.5$ shares is the average inventory per instrument between replenishment events. The resulting K-NPR capital charge (IFR Art. 22) is approximately 16 % of this notional.

A.2 Order Throughput Analysis

Figure A.1 presents the throughput surface across the full price range. The left panel renders the three-dimensional relationship between average share price, daily order flow and trades per second on a logarithmic price axis, with the capacity window of 10–20 trades per second shown as horizontal planes. The right panel projects the same surface as a heatmap, making the safe and breach zones immediately legible.

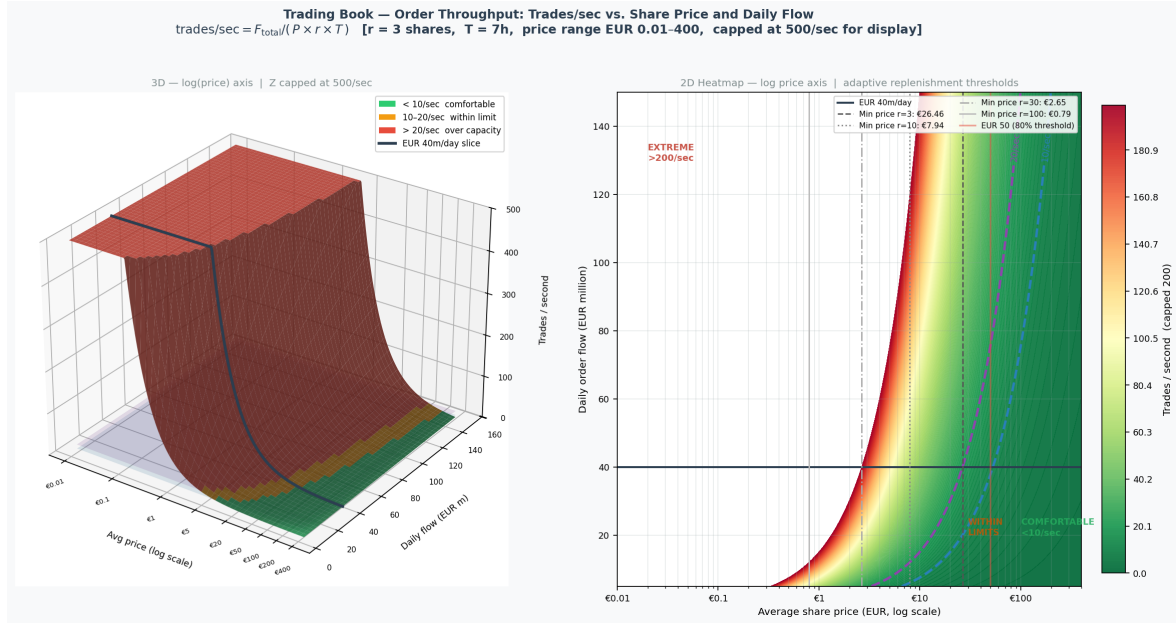


Figure A.1: Order throughput surface. Green = below 10 trades/sec (comfortable); amber = 10–20 (within capacity window); red = above 20 (over capacity). The logarithmic price axis spans €0.01 to €400. The vertical dashed lines in the right panel show the minimum average price required to remain within the 20 trades/sec limit for each replenishment window size r . The horizontal line marks the €40 m/day base-case flow.

The central finding is that with a fixed three-share replenishment window, stocks priced below approximately €26 breach the 20 trades/sec upper bound at €40 m/day flow. This is an absolute constraint that does not depend on the number of instruments. At €5/share, the implied rate exceeds 1,600 trades/sec — more than two orders of magnitude beyond the available capacity. Any instrument universe that includes penny stocks or micro-cap equities therefore requires **price-adaptive replenishment**: the inventory window must widen in proportion to the fall in price so that each replenishment event covers a larger share quantity. Table A.1 sets out the recommended tiers derived from equation (A.1) by solving for the minimum r that keeps trades/sec ≤ 20 at each price band.

Table A.1: Price-adaptive replenishment tiers — minimum replenishment size to remain within the 20 trades/sec capacity limit at €40 m/day flow

Price range	Rep. size r	Min avg price	Inventory window example
< €5 (micro-cap / penny)	500 shares	€0.16	min = 500, max = 1,000
€5–20 (small)	100 shares	€0.79	min = 100, max = 200
€20–50 (mid-small)	30 shares	€2.65	min = 30, max = 60
€50–150 (mid)	10 shares	€7.94	min = 10, max = 20
€150–400 (premium)	3 shares	€26.46	min = 2, max = 5

With price-adaptive replenishment in place, a realistic 80/20 universe of 5,000 instruments — 4,000 instruments at an average price of €25 and 1,000 at €200 — generates only 2.3 trades/sec against the €40 m/day flow level, well within capacity regardless of the flow split between the two price tiers. This result holds because the replenishment size for the cheap tier (30 shares per execution) reduces the event frequency by a factor of ten relative to the three-share default, and the total notional per execution event increases commensurately.

A.3 Inventory Size and Capital Implications

Figure A.2 analyses trading book size across the same parameter space. The relationship between inventory size, K-NPR capital and the fixed K-DTF charge is the key risk management question: as the universe expands into cheap, high-volume stocks, inventory notional grows even though individual positions are small in EUR terms, because there are many of them.

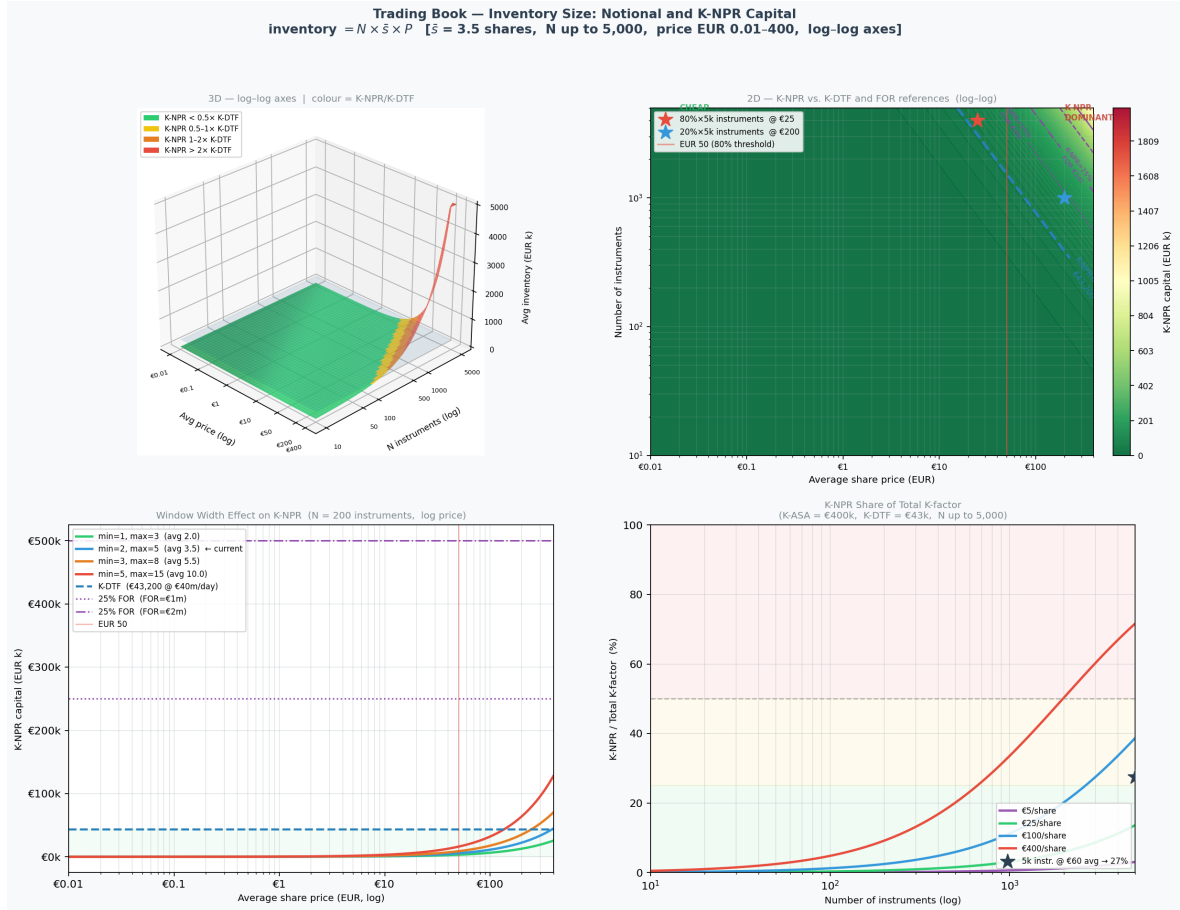


Figure A.2: Trading book inventory and K-NPR capital. *Top-left:* 3D surface of average inventory notional on log-log axes; colour indicates the ratio of K-NPR to K-DTF (green = K-NPR well below K-DTF; red = K-NPR dominant). *Top-right:* 2D heatmap of K-NPR capital; the blue dashed curve marks the K-NPR = K-DTF iso-line (EUR 43,200) and the purple contours show where K-NPR equals 25% of FOR at €0.5m, €1m, €2m and €3m annual fixed cost levels. The red and blue stars mark the two sub-populations of the realistic 80/20 universe (4,000 instruments at €25 and 1,000 at €200). *Bottom-left:* effect of inventory window width on K-NPR at $N = 200$ instruments across the full price range. *Bottom-right:* K-NPR as a share of total K-factor as instrument count grows; the star marks the realistic 5,000-instrument universe.

For the base-case parameters — inventory window $\text{min} = 2$, $\text{max} = 5$ (3.5 average shares held) — several important relationships emerge from Figure A.2. First, K-NPR remains below K-DTF across essentially the entire cheap and mid-price range at any reasonable instrument count. The K-NPR = K-DTF crossover (the blue dashed curve in the top-right panel) falls in the region of high price and large universe; reaching it requires either premium stocks above €200 or a universe of several thousand instruments. For the realistic 80/20 universe of 5,000 instruments at a weighted average price of €60, K-NPR reaches €168,000, compared to K-DTF of €43,200. K-NPR is therefore the larger K-factor component but remains modest in absolute terms, contributing approximately 27% of total K-factor at €1 bn of custody assets under management.

Second, the inventory window width is a material lever on K-NPR but not on throughput (since the replenishment size that solves the throughput constraint simultaneously determines the average inventory). The bottom-left panel of Figure A.2 shows that widening from a $\min = 2/\max = 5$ window to $\min = 5/\max = 15$ roughly triples K-NPR at any given price level. For cheap stocks where a wide window is forced by the throughput constraint, this capital cost must be accepted; the only mitigation is to hold fewer instruments in that price tier or to exclude instruments below a chosen price floor altogether.

Third, the bottom-right panel of Figure A.2 makes visible the point at which the trading book starts to dominate the K-factor requirement. At €5/share (purple line), K-NPR never exceeds 10% of total K-factor even at 5,000 instruments. At €25/share (green), K-NPR reaches approximately 22% at 5,000 instruments. At €100/share (blue), it crosses 50% at around 1,200 instruments. At €400/share (red), it dominates from 500 instruments upward. The practical implication is that instrument eligibility criteria should be set with reference to this crossover point: admitting large numbers of high-priced equities into the universe materially shifts the binding own funds constraint from K-DTF (driven by flow) toward K-NPR (driven by inventory), which is harder to reduce without restructuring the inventory thresholds or reducing the universe.

A.4 Design Recommendations from the Quantitative Analysis

The quantitative analysis supports three concrete design recommendations.

The first is the adoption of **price-adaptive inventory thresholds** from the outset. Applying a uniform $\min = 2/\max = 5$ window across all price tiers is operationally unsustainable for instruments below €50: the resulting execution frequency would immediately saturate the available order throughput. The tiers in Table A.1 provide a proportionate mapping between price level and replenishment size that maintains throughput within the 10–20 trades/sec capacity window across the full instrument universe.

The second is a **minimum share price floor** on instrument eligibility. Instruments priced below approximately €1 generate throughput requirements that cannot be managed even with a 500-share replenishment window at €40 m/day flow, and they contribute negligible inventory notional. A minimum price floor of €1 to €5, combined with the adaptive replenishment tiers above, eliminates the problematic tail of the distribution while preserving a broad accessible universe.

The third is the recognition that **K-DTF, not K-NPR, is the primary K-factor driver at the envisaged flow level and instrument universe**. K-DTF is fixed at €43,200 regardless of instrument count and price; it scales only with total daily fractional order flow. K-NPR at the realistic 5,000- instrument universe with a weighted

average price of €60 reaches €168,000 — roughly four times K-DTF. Both are modest relative to the Fixed Overhead Requirement, which at €10m annual fixed costs stands at €2.5m, confirming that FOR will remain the binding own funds pillar through the foreseeable growth trajectory of the business.